Seminar 5 - Optimization

ECE 447: Data Analysis and Machine Learning for Engineers

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Why **Optimization**?

Optimization is the backbone of almost all concepts in machine learning.

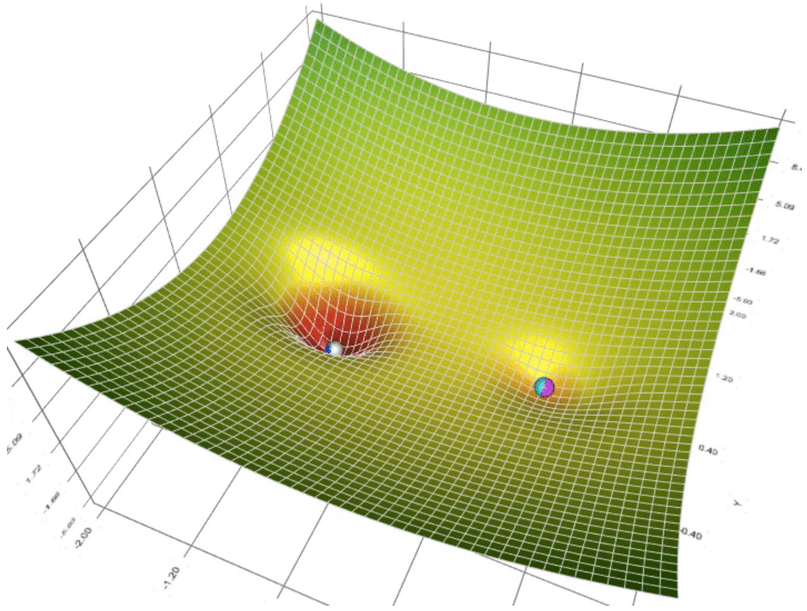
As engineers, we formulate conceptual ideas into optimization problems:

$$\min_{x \in D} \qquad f(x)$$

What **algorithms** we should use to solve the problem?

Today's topic: **Gradient Descent (GD)**.

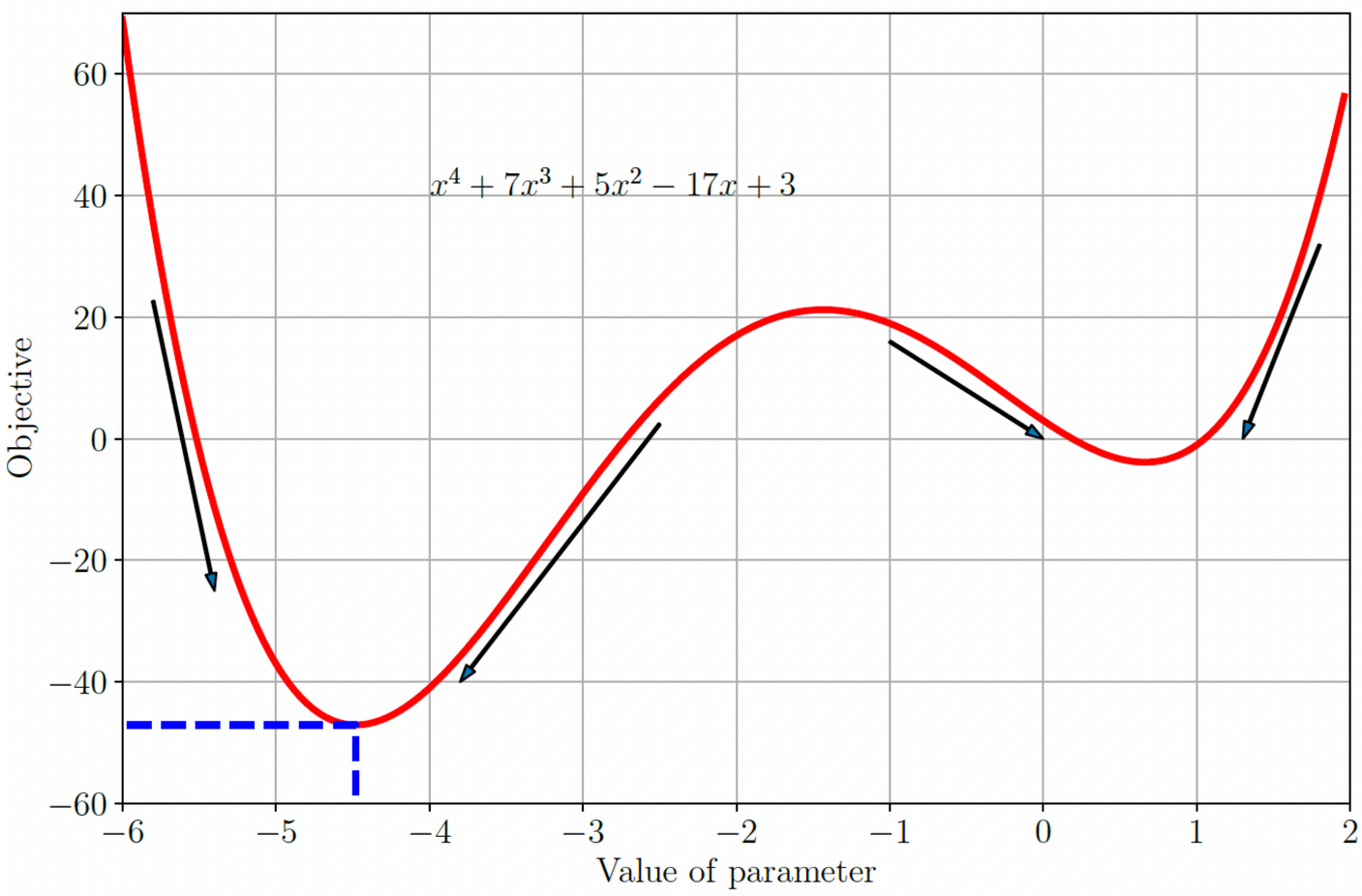
What is the underlying optimization behind gradient descent?



https://towardsdatascience.com.

Goal: Find the Minimum of the Function.

Move in the direction that decreases the function value.



This Seminar Covers ...

- How to Approximate a Complicated Function?
- Introduction to Convex Optimization
- Gradient Descent Formulation
- Optimization Interpretation of Gradient Descent
- Beyond Gradient Descent

How to Approximate a Complicated Function?

Taylor Series Expansion

$$f(x) = f(x_0) + \sum_{k=1}^{\infty} \frac{1}{k!} \frac{\partial^k f(x_0)}{\partial x^k} (x - x_0)^k$$

First order expansion:

$$f(x) = f(x_0) + \frac{\partial f(x_0)}{\partial x}(x - x_0) + \text{error}$$

Approximate with a line

Second order expansion:

$$f(x) = f(x_0) + \frac{\partial f(x_0)}{\partial x}(x - x_0) + \frac{1}{2}\frac{\partial^2 f(x_0)}{\partial x^2}(x - x_0)^2 + \text{error}$$

Quadratic approximation

Second Taylor Series Expansion for Vectors

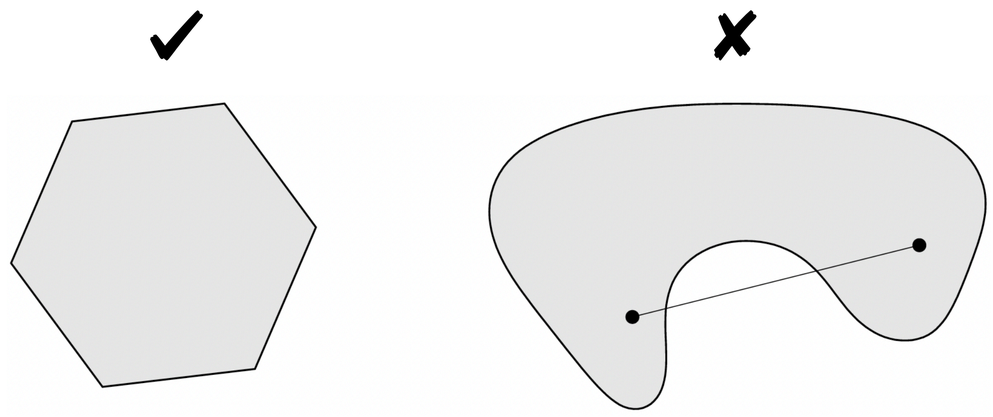
For a function $f : \mathbb{R}^n \to \mathbb{R}$, and a vector $\boldsymbol{x} = [x_1, \ldots, x_n]^\top \in \mathbb{R}^n$, the second order Taylor approximation around $\boldsymbol{x}_0$ is

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)^\top (\boldsymbol{x} - \boldsymbol{x}_0) + \frac{1}{2} (\boldsymbol{x} - \boldsymbol{x}_0)^\top \nabla^2 f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0)$$

Lets see a desmos visualization together.

Convex Sets

A set is *convex*, if the line segment between any two elements lies entirely in the set.



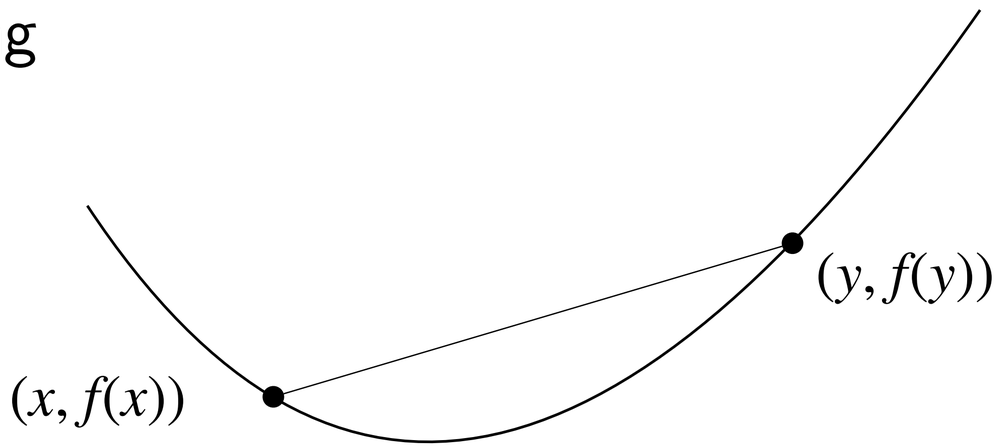
A set $C \subseteq \mathbb{R}^n$ is *convex* if

for any $x, y \in C$ and any θ with $0 \leq \theta \leq 1$, we have

$$\theta x + (1 - \theta) y \in C.$$

Convex Functions

A function is *convex* if the line segment joining $f(x)$ and $f(y)$ lies above the graph of f.



domain *image*

A function $f : \mathbb{R}^n \to \mathbb{R}$ is *convex* if $\mathrm{dom}(f)$ is a convex set and if

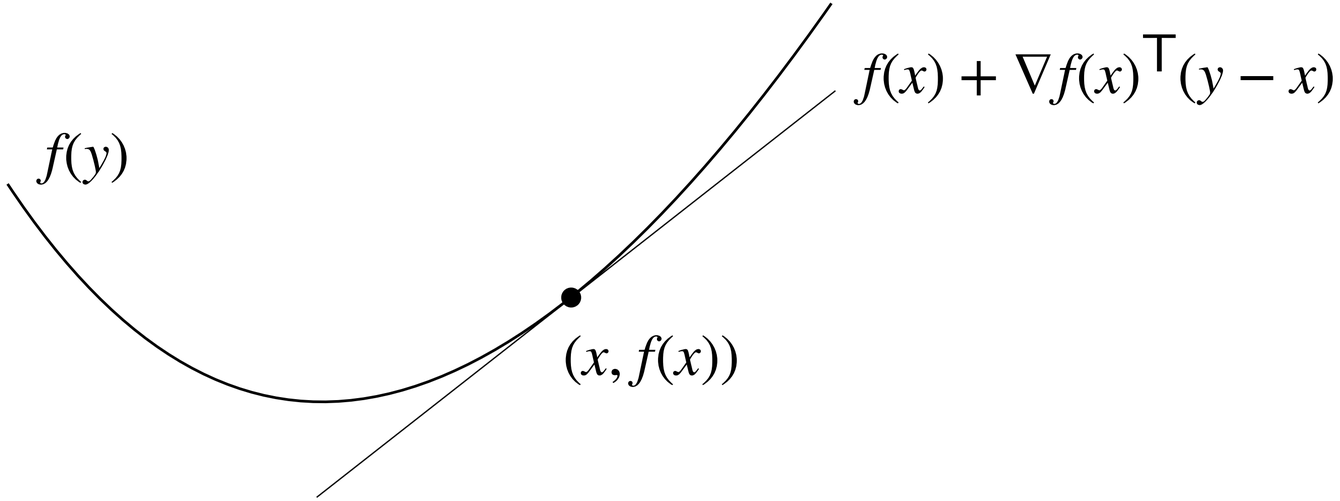
for all $\boldsymbol{x}, \boldsymbol{y} \in \mathrm{dom}(f)$, and θ with $0 \leq \theta \leq 1$, we have

$$f(\theta \boldsymbol{x} + (1-\theta)\boldsymbol{y}) \quad \leq \quad \theta f(\boldsymbol{x}) + (1-\theta) f(\boldsymbol{y}).$$

First-Order Conditions of Convex Functions

Suppose f is differentiable. Then f is convex if and only if $\mathrm{dom}(f)$ is convex and

$$f(y) \quad \geq \quad f(x) + \nabla f(x)^\top (y - x).$$



for all $y \in \mathrm{dom}(f)$, $\nabla f(x) = 0 \quad \Leftrightarrow \quad f(y) \geq f(x)$ *(x is a global minimizer of f)*

Second-Order Conditions of Convex Functions

Suppose f is twice-differentiable. Then f is convex if and only if $\mathrm{dom}(f)$ is convex and

$$\nabla^2 f(x) \succeq 0 \text{ for all } x \in \mathrm{dom}(f).$$

Hessian

The graph of the function have positive (upward) curvature.

Examples:

- *(Exponentials)* e^{ax}
- *(Powers)* x^2
- *(Norms)* $\|\boldsymbol{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_n^2$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Optimization Terminology

Convex optimization problem:

$$\begin{aligned} \min_{x \in D} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m \\ & Ax = b \end{aligned}$$

- f is called criterion, objective, or loss function.
- The set of points that $x \in D$, $g_i(x) \leq 0$, and $Ax = b$ is a feasible set.
- The minimum of $f(x)$ over all feasible points x is called optimal value, denoted as f^*.
- If x is feasible and $f(x) = f^*$, then x is optimal, solution, or minimizer.

(if there exists a solution)

Gradient Descent Formulation

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex and differentiate function.

Given the unconstrained, smooth convex optimization $\min_{x \in \mathbb{R}^n} \quad f(x)$

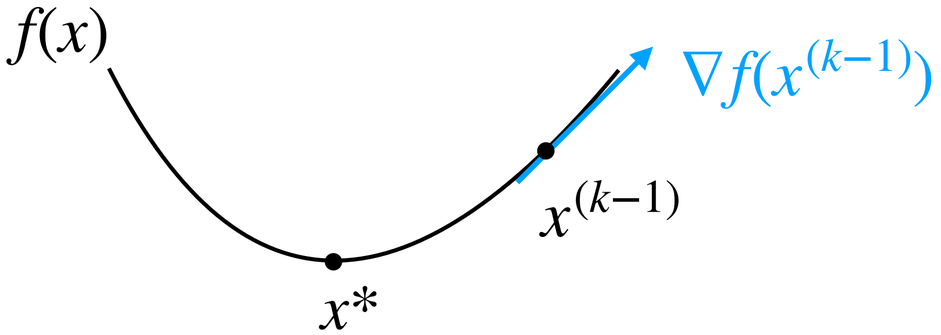
find x^*, the minimizer of $f(x)$.

$$x^* = \arg\min_{x \in \mathbb{R}^n} \quad f(x)$$

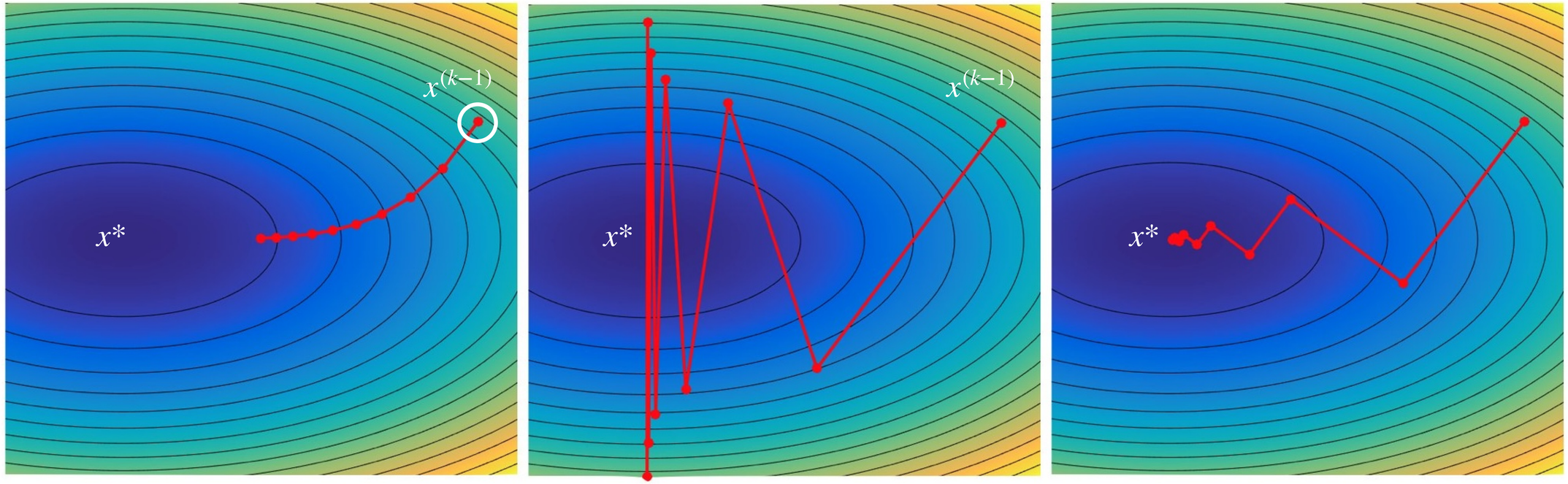
Gradient Descent: find x^*, using iterative update with initial $x^{(0)}$ until some convergence.

$$x^{(k)} = x^{(k-1)} - \eta \nabla f(x^{(k-1)})$$

learning rate (step size)



Loss Landscape and Learning Rate of Gradient Descent



Small step size | Large step size | Suitable step size

Image from Gabriel Peyré's twitter.

The Optimization Behind Gradient Descent: Majorization Minimization

What is the local objective $h(x; x^{(k-1)})$ that gradient descent optimizes at each step?

$$x^{(k)} = x^{(k-1)} - \eta \nabla f(x^{(k-1)}) \quad = \arg\min_{x \in \mathbb{R}^n} \; h(x; x^{(k-1)})$$

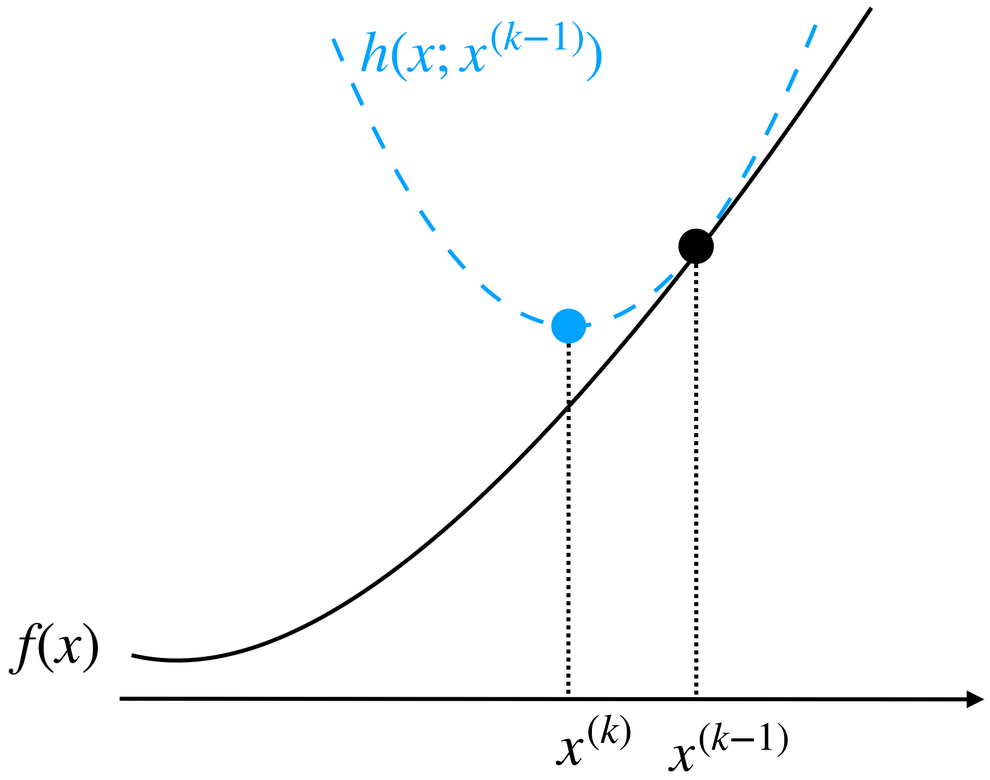
Majorization Minimization (MM):

Define a simpler surrogate function,

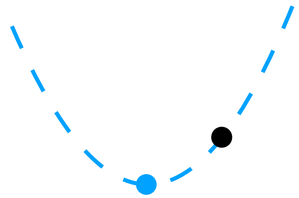
and minimize iteratively.

$$h(x; x^{(k-1)}) \;\geq\; f(x) \quad \text{for all } x,$$

$$h(x^{(k-1)}; x^{(k-1)}) \;=\; f(x^{(k-1)}).$$



Gradient Descent as an MM Quadratic Approximation

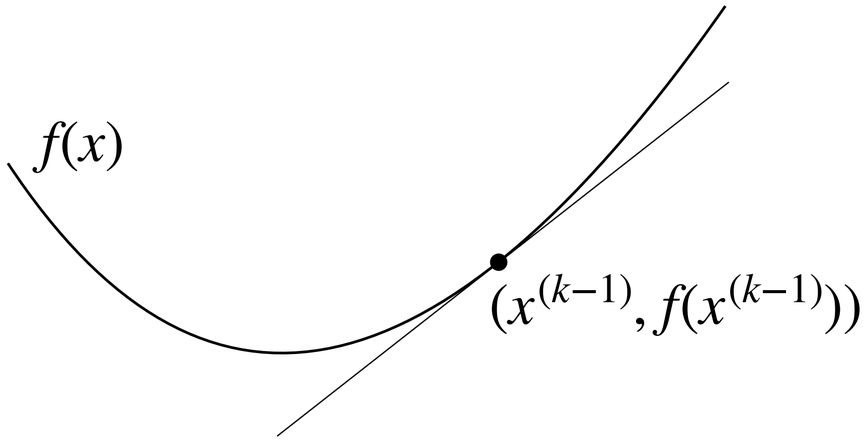


Choose next point $x^{(k)}$ to minimize an MM quadratic approximation of the function.

$$x^{(k)} = \arg\min_{x \in \mathbb{R}^n} \quad f(x^{(k-1)}) + \nabla f(x^{(k-1)})^\top (x - x^{(k-1)}) + \frac{1}{2\eta} \|(x - x^{(k-1)})\|_2^2$$

Linear approximation to f

$$f(x^{(k-1)}) + \nabla f(x^{(k-1)})^\top (x - x^{(k-1)})$$



Proximity term to $x^{(k-1)}$

$$\frac{1}{2\eta} \|(x - x^{(k-1)})\|_2^2$$

Large regularization: staying close to $x^{(k-1)}$

(small step size η)

Small regularization: can get far from $x^{(k-1)}$

(large step size η)

Proof

$$\min_x \quad f(x^{(k-1)}) + \nabla f(x^{(k-1)})^\top (x - x^{(k-1)}) + \frac{1}{2\eta}\|(x - x^{(k-1)})\|_2^2$$

$$= \min_x \quad \nabla f(x^{(k-1)})^\top x - \nabla f(x^{(k-1)})^\top x^{(k-1)} + \frac{1}{2\eta}\|(x - x^{(k-1)})\|_2^2 \qquad \text{(remove constant)}$$

$$= \min_x \quad 2\eta\, \nabla f(x^{(k-1)})^\top x - 2\eta\, \nabla f(x^{(k-1)})^\top x^{(k-1)} + \|(x - x^{(k-1)})\|_2^2 \qquad \text{(scale by } 2\eta\text{)}$$

$$= \min_x \quad 2\eta\, \nabla f(x^{(k-1)})^\top x - 2\eta\, \nabla f(x^{(k-1)})^\top x^{(k-1)} + \|x\|_2^2 - 2x^\top x^{(k-1)} + \|x^{(k-1)}\|_2^2 \qquad \text{(expand the second term)}$$

(add one constant)

$$= \min_x \quad 2\eta\, \nabla f(x^{(k-1)})^\top x - 2\eta\, \nabla f(x^{(k-1)})^\top x^{(k-1)} + \|x\|_2^2 - 2x^\top x^{(k-1)} + \|x^{(k-1)}\|_2^2 + \|\eta\, \nabla f(x^{(k-1)})\|_2^2$$

$$= \min_x \quad \underbrace{\|x\|_2^2}_{\text{first term}} - 2x^\top (x^{(k-1)} - \eta\, \nabla f(x^{(k-1)})) + \underbrace{\|x^{(k-1)} - \eta\, \nabla f(x^{(k-1)})\|_2^2}_{\text{second term}} \qquad \text{(re-arrange and combine)}$$

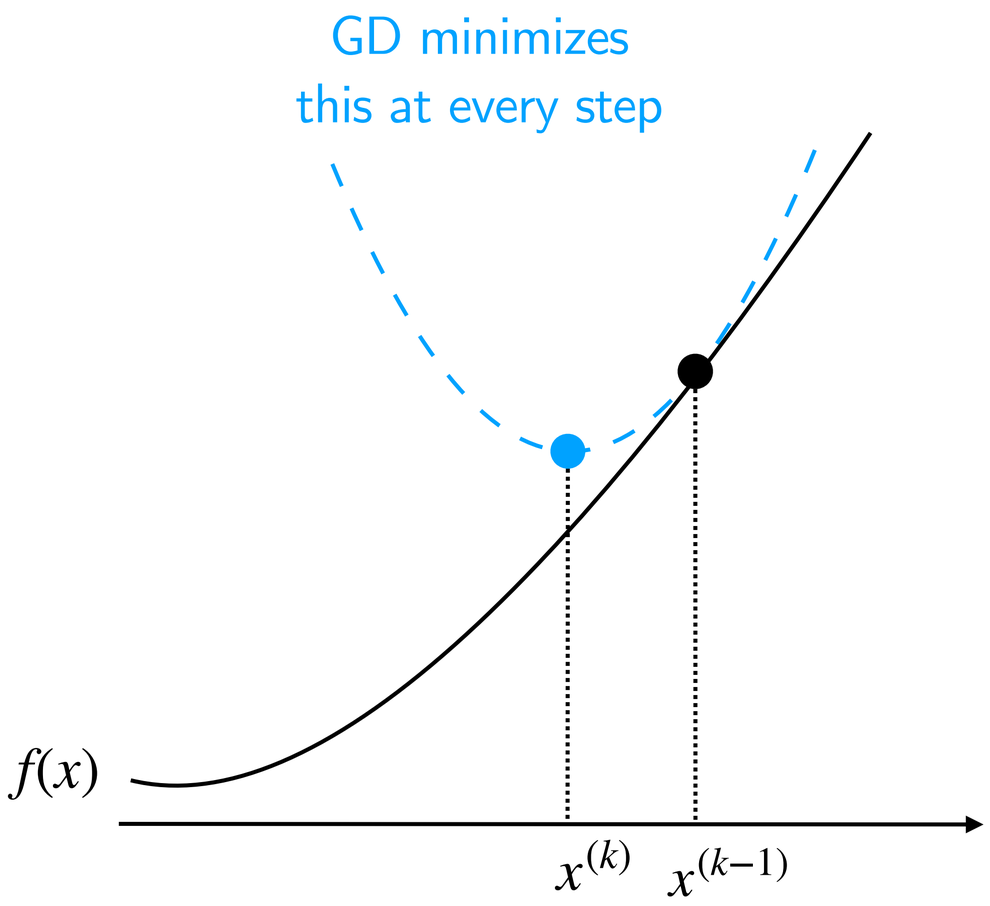
$$= \min_x \|x - [x^{(k-1)} - \eta\, \nabla f(x^{(k-1)})]\|_2^2 \qquad \Rightarrow \qquad x = x^{(k-1)} - \eta\, \nabla f(x^{(k-1)})$$

What Happened?

We showed that gradient descent is an MM, i.e.,

Gradient descent with proper step size (learning rate) can find a minimizer of a convex function (loss).

Gradient descent is an iterative algorithm to minimize a loss function.



Beyond Gradient Descent

Proximal Gradient Descent

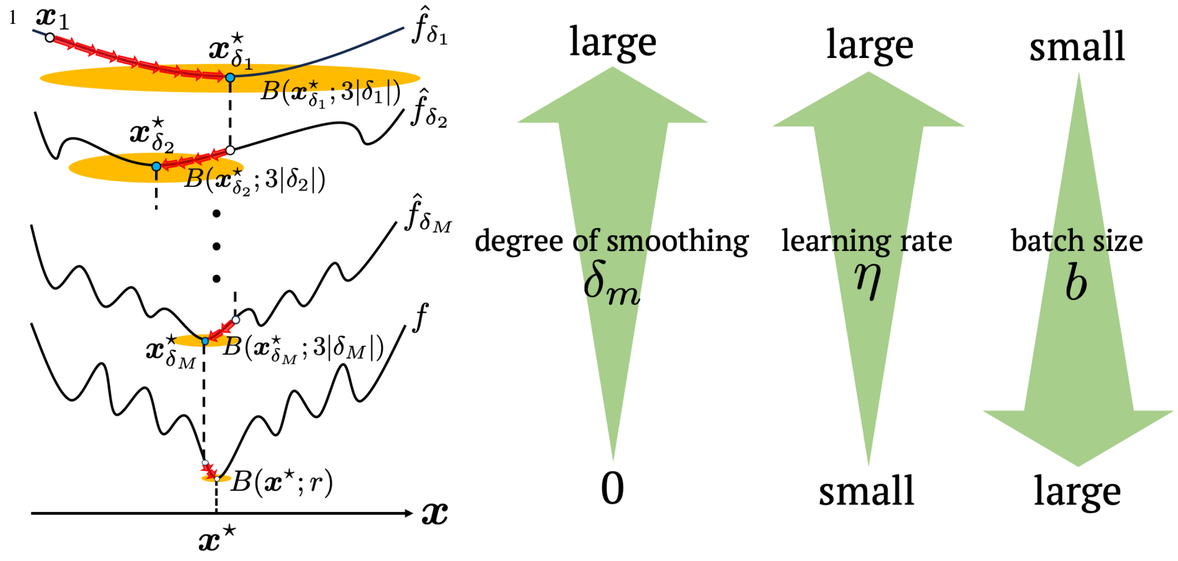
$$\min_{x \in \mathbb{R}^n} \quad g(x) + h(x)$$

g is convex, differentiable.

h is convex, not necessarily differentiable.

$$x^{(k)} = \text{prox}_{h,\eta}(x^{(k-1)} - \eta \nabla g(x^{(k-1)}))$$

Stochastic Gradient Descent

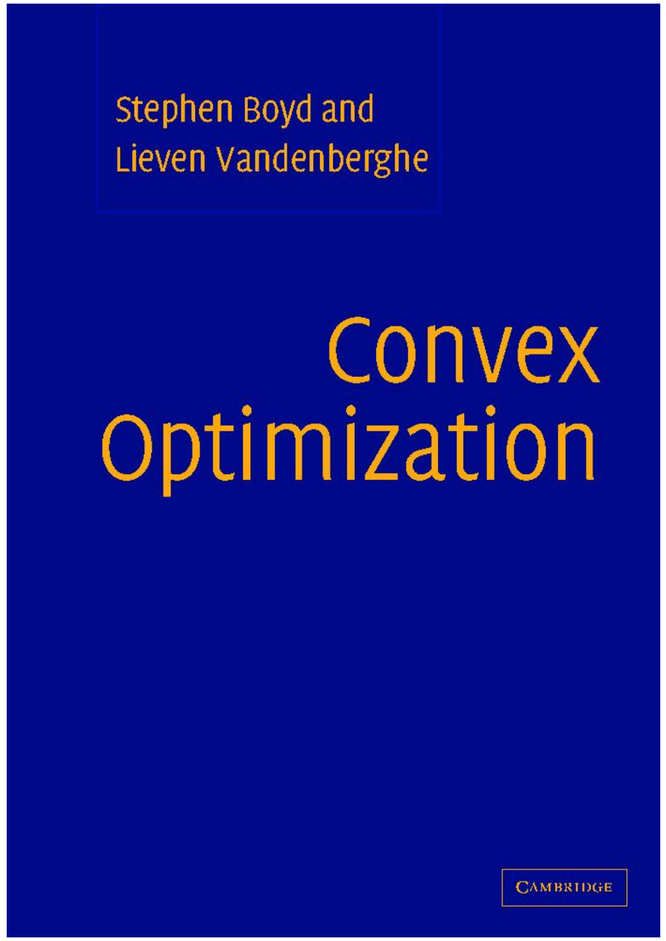


$$\min_{x \in \mathbb{R}^n} \quad \frac{1}{m} \sum_{i=1}^{m} f_i(x)$$

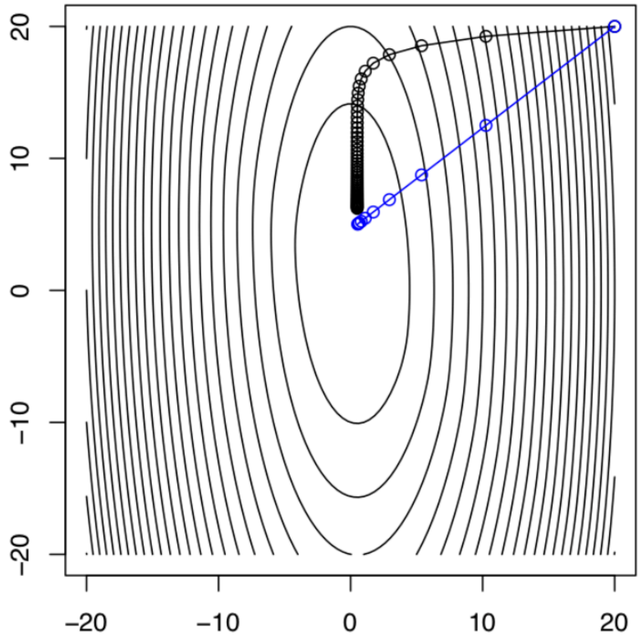
$$x^{(k)} = (x^{(k-1)} - \eta \nabla f_i(x^{(k-1)}))$$

[1]Sato, N., & Iiduka, H. (2023). Using Stochastic Gradient Descent to Smooth Nonconvex Functions: Analysis of Implicit Graduated Optimization with Optimal Noise Scheduling. *arXiv.*

References for Students Interested in Optimization



Ryan Tibshirani - Convex Optimization at CMU



$$x^+ = x - t\nabla f(x)$$

$$x^+ = x - (\nabla^2 f(x))^{-1}\nabla f(x)$$